

89

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QUANT CHECKLIST

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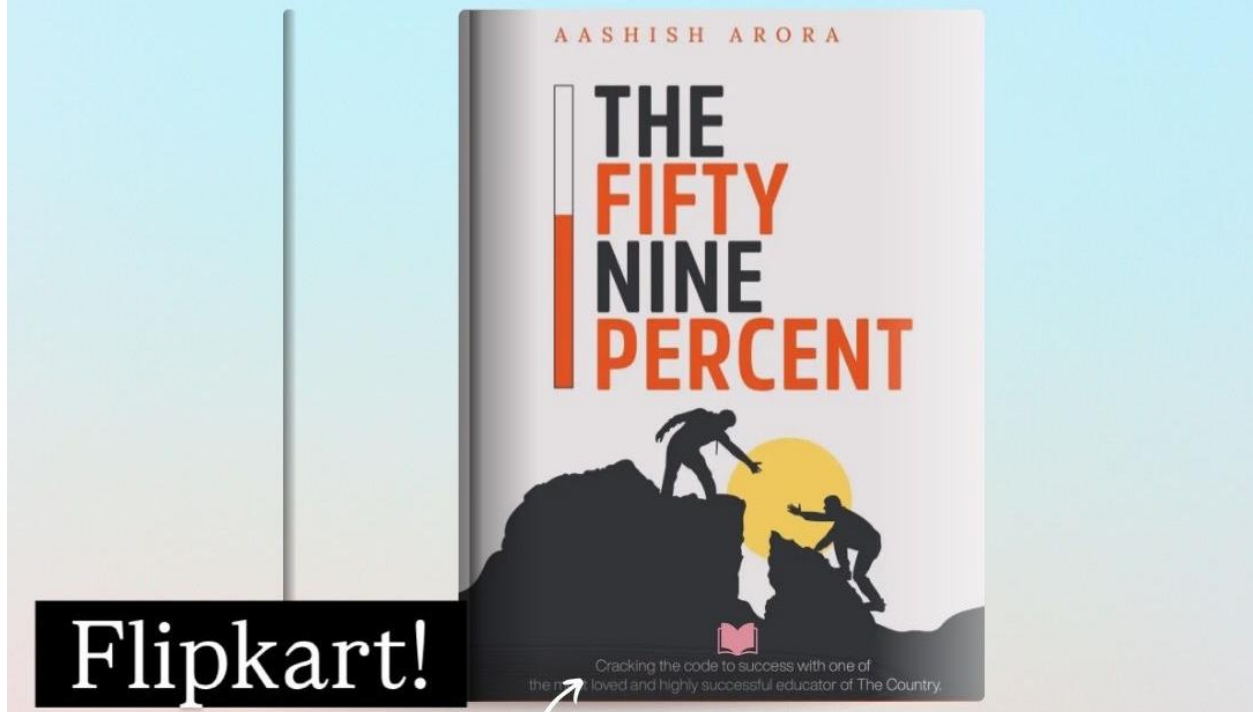


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1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $37.5\% \text{ of } 864 + \sqrt{5184} - \frac{2}{11} \text{ of } 363 = ? \times 6 + 45$

- A. 45.5
- B. 47.5
- C. 49.5
- D. 51.5
- E. None of these

2. $? + 24 \times 18 = 83.33\% \text{ of } 972 + \sqrt[3]{91125}$

- A. 413
- B. 417
- C. 423
- D. 429
- E. None of these

3. $62.5\% \text{ of } 448 + 13^2 - \frac{7}{12} \text{ of } 360 = ? + 117$

- A. 118
- B. 120
- C. 122

D. 126

- E. None of these

4. $(\sqrt{6084} + \sqrt[3]{2744}) \times 7 - 41.66\% \text{ of } 1440 = ? \times 5$

- A. 7.8
- B. 8.2
- C. 8.8
- D. 9.4
- E. None of these

5. $4\frac{1}{5} \text{ of } 150 - 2\frac{3}{4} \text{ of } 128 + 37.5\% \text{ of } 384 = ? + 221$

- A. 191
- B. 197
- C. 201
- D. 207
- E. None of these

6. $?^2 + 56.25\% \text{ of } 576 = 22 \times 32 - \sqrt{3136}$

- A. 14
- B. 16
- C. 18

D. 20

E. None of these

7. 6.25% of 4096 + 31.25% of 640 + 14.28% of 700 = ? $\times 8 - 64$

A. 72.5

B. 75.5

C. 77.5

D. 80.5

E. None of these

8. $(16^3 \div 32) + (27^3 \div 81) - 68.75\%$ of 320 = ? + 65

A. 76

B. 82

C. 86

D. 90

E. None of these

9. 44.44% of 729 + 27.27% of 1210 - $\frac{5}{7}$ of 490 = ? $\times 8 + 40$

A. 29

B. 31

C. 33

D. 35

E. None of these

10. 18.18% of 968 + $\frac{3}{8}$ of 1216 + $\sqrt{6561} = ? - 126$

A. 829

B. 835

C. 839

D. 845

E. None of these

11. $[\frac{3}{4} \text{ of } 512 + \frac{5}{8} \text{ of } 384] \div ? = \sqrt[3]{5832} + 60$

A. 6

B. 7

C. 8

D. 9

E. None of these

12. 58.33% of 1728 - 71.42% of 1617 + 125% of 496 = ? + 432

A. 37

B. 39

C. 41

D. 43

E. None of these

13. $\sqrt{?} - 18.75\%$ of 768 = $16^2 - 12^2$

A. 62500

B. 65536

C. 67600

D. 70756

E. None of these

14. $28 \times 36 - 16 \times 42 + 75\% \text{ of } 864 = ?$
 $\times 12$

A. 78

B. 80

C. 82

D. 84

E. None of these

15. $9\frac{1}{6} + 7\frac{5}{12} - 3\frac{3}{4} + 4\frac{1}{6} = ?$

A. 15

B. 16

C. 17

D. 18

E. None of these

16. $? \times 11 + 31.25\% \text{ of } 512 = \sqrt[3]{175616}$
 $+ 18^2$

A. 18

B. 20

C. 22

D. 24

E. None of these

17. $(5^2 \times 4^3) \div 40 + 83.33\% \text{ of } 288 -$
 $6.25\% \text{ of } 1600 = ?$

A. 170

B. 175

C. 180

D. 185

E. None of these

18. $225\% \text{ of } 320 - 175\% \text{ of } 264 +$
 $68.75\% \text{ of } 448 = ? + 200$

A. 356

B. 362

C. 366

D. 372

E. None of these

19. $? \div 9 + \sqrt{3600} = 37.5\% \text{ of } 720 + 14$
 $\times 18$

A. 4058

B. 4158

C. 4258

D. 4358

E. None of these

20. $[(46^2 - 34^2) \div 16] + \frac{3}{7} \text{ of } 392 + 2^3 =$
 $? \times 4$

A. 55

B. 57

C. 59

D. 61

E. None of these

Answers:

1. B

2. C

3. C

4. C

5. C

6. C

7. C

8. C

9. C

10. C

11. C

12. C

13. B

14. C

15. C

16. B

17. C

18. C

19. B

20. C

Solutions:

$$1. 37.5\% \text{ of } 864 + \sqrt{5184} - \frac{2}{11} \text{ of } 363 = ? \times 6 + 45$$

$$37.5\% = \frac{3}{8}, \sqrt{5184} = 72, \text{ and } \frac{2}{11} \text{ of } 363 = 66$$

$$\frac{3}{8} \times 864 + 72 - 66 = ? \times 6 + 45$$

$$324 + 72 - 66 = ? \times 6 + 45$$

$$330 = ? \times 6 + 45$$

$$? \times 6 = 330 - 45 = 285$$

$$? = 285 \div 6 = 47.5$$

$$2. ? + 24 \times 18 = 83.33\% \text{ of } 972 + \sqrt[3]{91125}$$

$$83.33\% = \frac{5}{6}, 24 \times 18 = 432, \text{ and } \sqrt[3]{91125} = 45$$

$$? + 432 = \frac{5}{6} \times 972 + 45$$

$$? + 432 = 810 + 45$$

$$? + 432 = 855$$

$$? = 855 - 432 = 423$$

$$3. 62.5\% \text{ of } 448 + 13^2 - \frac{7}{12} \text{ of } 360 = ? + 117$$

$$62.5\% = \frac{5}{8}, 13^2 = 169, \text{ and } \frac{7}{12} \text{ of } 360 = 210$$

$$\frac{5}{8} \times 448 + 169 - 210 = ? + 117$$

$$280 + 169 - 210 = ? + 117$$

$$239 = ? + 117$$

$$? = 239 - 117 = 122$$

$$4. (\sqrt{6084} + \sqrt[3]{2744}) \times 7 - 41.66\% \text{ of } 1440 = ? \times 5$$

$$\sqrt{6084} = 78, \sqrt[3]{2744} = 14, \text{ and } 41.66\% = \frac{5}{12}$$

$$(78 + 14) \times 7 - \frac{5}{12} \times 1440 = ? \times 5$$

$$92 \times 7 - 600 = ? \times 5$$

$$644 - 600 = ? \times 5$$

$$44 = ? \times 5$$

$$? = 44 \div 5 = 8.8$$

$$5. 4\frac{1}{5} \text{ of } 150 - 2\frac{3}{4} \text{ of } 128 + 37.5\% \text{ of } 384 = ? + 221$$

$$4\frac{1}{5} = \frac{21}{5}, 2\frac{3}{4} = \frac{11}{4}, \text{ and } 37.5\% = \frac{3}{8}$$

$$\frac{21}{5} \times 150 - \frac{11}{4} \times 128 + \frac{3}{8} \times 384 = ? + 221$$

$$630 - 352 + 144 = ? + 221$$

$$422 = ? + 221$$

$$? = 422 - 221 = 201$$

$$6. ?^2 + 56.25\% \text{ of } 576 = 22 \times 32 - \sqrt{3136}$$

$$56.25\% = \frac{9}{16}, 22 \times 32 = 704, \text{ and } \sqrt{3136} = 56$$

$$?^2 + \frac{9}{16} \times 576 = 704 - 56$$

$$?^2 + 324 = 648$$

$$?^2 = 648 - 324 = 324$$

$$? = \sqrt{324} = 18$$

$$7. 6.25\% \text{ of } 4096 + 31.25\% \text{ of } 640 + 14.28\% \text{ of } 700 = ? \times 8 - 64$$

$$6.25\% = \frac{1}{16}, 31.25\% = \frac{5}{16}, \text{ and } 14.28\% = \frac{1}{7}$$

$$\frac{1}{16} \times 4096 + \frac{5}{16} \times 640 + \frac{1}{7} \times 700 = ? \times 8 - 64$$

$$256 + 200 + 100 = ? \times 8 - 64$$

$$556 = ? \times 8 - 64$$

$$? \times 8 = 556 + 64 = 620$$

$$? = 620 \div 8 = 77.5$$

$$8. (16^3 \div 32) + (27^3 \div 81) - 68.75\% \text{ of } 320 = ? + 65$$

$$16^3 = 4096, 27^3 = 19683, \text{ and } 68.75\% = \frac{11}{16}$$

$$4096 \div 32 + 19683 \div 81 - \frac{11}{16} \times 320 = ? + 65$$

$$128 + 243 - 220 = ? + 65$$

$$151 = ? + 65$$

$$? = 151 - 65 = 86$$

$$9. 44.44\% \text{ of } 729 + 27.27\% \text{ of } 1210 - \frac{5}{7} \text{ of } 490 = ? \times 8 + 40$$

$$44.44\% = \frac{4}{9}, 27.27\% = \frac{3}{11}, \text{ and } \frac{5}{7} \text{ of } 490 = 350$$

$$\frac{4}{9} \times 729 + \frac{3}{11} \times 1210 - 350 = ? \times 8 + 40$$

$$324 + 330 - 350 = ? \times 8 + 40$$

$$304 = ? \times 8 + 40$$

$$? \times 8 = 304 - 40 = 264$$

$$? = 264 \div 8 = 33$$

$$10. 18.18\% \text{ of } 968 + \frac{3}{8} \text{ of } 1216 + \sqrt{6561} = ? - 126$$

$$18.18\% = \frac{2}{11}, \sqrt{6561} = 81, \text{ and } \frac{3}{8} \text{ of } 1216 = 456$$

$$\frac{2}{11} \times 968 + 456 + 81 = ? - 126$$

$$176 + 456 + 81 = ? - 126$$

$$713 = ? - 126$$

$$? = 713 + 126 = 839$$

$$11. \left[\frac{3}{4} \text{ of } 512 + \frac{5}{8} \text{ of } 384 \right] \div ? = \sqrt[3]{5832} + 60$$

$$\frac{3}{4} \text{ of } 512 = 384, \frac{5}{8} \text{ of } 384 = 240, \text{ and } \sqrt[3]{5832} = 18$$

$$(384 + 240) \div ? = 18 + 60$$

$$624 \div ? = 78$$

$$? = 624 \div 78 = 8$$

$$12. 58.33\% \text{ of } 1728 - 71.42\% \text{ of } 1617 + 125\% \text{ of } 496 = ? + 432$$

$$58.33\% = \frac{7}{12}, 71.42\% = \frac{5}{7}, \text{ and } 125\% = \frac{5}{4}$$

$$\frac{7}{12} \times 1728 - \frac{5}{7} \times 1617 + \frac{5}{4} \times 496 = ? + 432$$

$$1008 - 1155 + 620 = ? + 432$$

$$473 = ? + 432$$

$$? = 473 - 432 = 41$$

$$13. \sqrt{?} - 18.75\% \text{ of } 768 = 16^2 - 12^2$$

$$18.75\% = \frac{3}{16}, 16^2 = 256, \text{ and } 12^2 = 144$$

$$\sqrt{?} - \frac{3}{16} \times 768 = 256 - 144$$

$$\sqrt{?} - 144 = 112$$

$$\sqrt{?} = 112 + 144 = 256$$

$$? = 256^2 = 65536$$

$$14. 28 \times 36 - 16 \times 42 + 75\% \text{ of } 864 = ? \times 12$$

$$75\% = \frac{3}{4}, 28 \times 36 = 1008, \text{ and } 16 \times 42 = 672$$

$$1008 - 672 + \frac{3}{4} \times 864 = ? \times 12$$

$$1008 - 672 + 648 = ? \times 12$$

$$984 = ? \times 12$$

$$? = 984 \div 12 = 82$$

$$15. 9\frac{1}{6} + 7\frac{5}{12} - 3\frac{3}{4} + 4\frac{1}{6} = ?$$

$$9\frac{1}{6} = \frac{55}{6}, 7\frac{5}{12} = \frac{89}{12}, 3\frac{3}{4} = \frac{15}{4}, \text{ and } 4\frac{1}{6} = \frac{25}{6}$$

$$\frac{55}{6} + \frac{89}{12} - \frac{15}{4} + \frac{25}{6} = ?$$

$$\frac{110}{12} + \frac{89}{12} - \frac{45}{12} + \frac{50}{12} = ?$$

$$\frac{204}{12} = ?$$

$$? = 17$$

$$16. ? \times 11 + 31.25\% \text{ of } 512 = \sqrt[3]{175616} + 18^2$$

$$31.25\% = \frac{5}{16}, \sqrt[3]{175616} = 56, \text{ and } 18^2 = 324$$

$$? \times 11 + \frac{5}{16} \times 512 = 56 + 324$$

$$? \times 11 + 160 = 380$$

$$? \times 11 = 380 - 160 = 220$$

$$? = 220 \div 11 = 20$$

$$17. (5^2 \times 4^3) \div 40 + 83.33\% \text{ of } 288 - 6.25\% \text{ of } 1600 = ?$$

$$5^2 = 25, 4^3 = 64, 83.33\% = \frac{5}{6}, \text{ and } 6.25\% = \frac{1}{16}$$

$$(25 \times 64) \div 40 + \frac{5}{6} \times 288 - \frac{1}{16} \times 1600 = ?$$

$$1600 \div 40 + 240 - 100 = ?$$

$$40 + 240 - 100 = ?$$

$$? = 180$$

$$18. 225\% \text{ of } 320 - 175\% \text{ of } 264 + 68.75\% \text{ of } 448 = ? + 200$$

$$225\% = \frac{9}{4}, 175\% = \frac{7}{4}, \text{ and } 68.75\% = \frac{11}{16}$$

$$\frac{9}{4} \times 320 - \frac{7}{4} \times 264 + \frac{11}{16} \times 448 = ? + 200$$

$$720 - 462 + 308 = ? + 200$$

$$566 = ? + 200$$

$$? = 566 - 200 = 366$$

$$19. ? \div 9 + \sqrt{3600} = 37.5\% \text{ of } 720 + 14 \times 18$$

$$\sqrt{3600} = 60, 37.5\% = \frac{3}{8}, \text{ and } 14 \times 18 = 252$$

$$? \div 9 + 60 = \frac{3}{8} \times 720 + 252$$

$$? \div 9 + 60 = 270 + 252$$

$$? \div 9 + 60 = 522$$

$$? \div 9 = 522 - 60 = 462$$

$$? = 462 \times 9 = 4158$$

$$20. [(46^2 - 34^2) \div 16] + \frac{3}{7} \text{ of } 392 + 2^3 = ? \times 4$$

$$46^2 = 2116, 34^2 = 1156, \frac{3}{7} \text{ of } 392 = 168, \text{ and } 2^3 = 8$$

$$(2116 - 1156) \div 16 + 168 + 8 = ? \times 4$$

$$960 \div 16 + 168 + 8 = ? \times 4$$

$$60 + 168 + 8 = ? \times 4$$

$$236 = ? \times 4$$

$$? = 236 \div 4 = 59$$

2. ARITHMETIC QUESTIONS

Q1: The same principal of Rs. x is deposited in two different accounts. In the first account, it earns simple interest at 15% p.a. for 4 years. In the second account, it earns compound interest at 10% p.a., compounded annually, for 2 years. If the sum of the amounts received from both accounts is Rs. 8430, find x .

समान मूलधन x रुपये दो अलग-अलग खातों में जमा किया गया। पहले खाते में 4 वर्षों के लिए 15% वार्षिक दर से साधारण ब्याज मिलता है। दूसरे खाते में 2 वर्षों के लिए 10% वार्षिक दर से चक्रवृद्धि ब्याज मिलता है, जो वार्षिक रूप से संयोजित होता है। यदि दोनों खातों से प्राप्त राशियों का योग 8430 रुपये है, तो x ज्ञात कीजिए।

- (A) Rs. 2500
- (B) Rs. 2800
- (C) Rs. 3200
- (D) Rs. 3000
- (E) Rs. 3500

Q2: The radius and height of a cylindrical container are in the ratio 2:3. If the volume of the container is 12936 cm^3 , then find its curved surface area. Take $\pi = 22/7$.

एक बेलनाकार पात्र की त्रिज्या और ऊँचाई का अनुपात 2:3 है। यदि पात्र का आयतन 12936 घन सेमी है, तो उसका वक्र पृष्ठीय क्षेत्रफल ज्ञात कीजिए। $\pi = 22/7$ मानिए।

- (A) 1764 cm^2
- (B) 1512 cm^2
- (C) 1936 cm^2
- (D) 1680 cm^2
- (E) 1848 cm^2

Q3: A train runs at a speed of 72 km/hr without any stoppage. During a journey, it stops 10 times and each stoppage is for 3 minutes. If the total journey time including stoppages is 4 hours 15 minutes, then find the distance covered by the train.

एक ट्रेन बिना रुके 72 किमी/घंटा की चाल से चलती है। यात्रा के दौरान वह 10 बार रुकती है और प्रत्येक ठहराव 3 मिनट का है। यदि ठहराव सहित कुल यात्रा समय 4 घंटे 15 मिनट है, तो ट्रेन द्वारा तय की गई दूरी ज्ञात कीजिए।

- (A) 300 km

- (B) 240 km
- (C) 270 km
- (D) 288 km
- (E) 260 km

Q4: A boat travels upstream from one point to another at a speed of 16 km/hr and returns the same distance downstream. If the ratio of time taken in upstream to downstream is 3:2, then find the speed of the stream.

एक नाव एक बिंदु से दूसरे बिंदु तक धारा के विपरीत 16 किमी/घंटा की चाल से जाती है और समान दूरी धारा के अनुकूल वापस आती है। यदि धारा के विपरीत और धारा के अनुकूल लगे समय का अनुपात 3:2 है, तो धारा की चाल ज्ञात कीजिए।

- (A) 5 km/hr
- (B) 4 km/hr
- (C) 6 km/hr
- (D) 8 km/hr
- (E) 3 km/hr

Q5: The average number of readers visiting a library in a week is 28. The average number of readers visiting on Monday, Tuesday and Wednesday is 32, while the average number of readers visiting on Friday, Saturday and Sunday is 25. Find the number of readers who visited the library on Thursday.

एक सप्ताह में पुस्तकालय आने वाले पाठकों की औसत संख्या 28 है। सोमवार, मंगलवार और बुधवार को आने वाले पाठकों की औसत संख्या 32 है, जबकि शुक्रवार, शनिवार और रविवार को आने वाले पाठकों की औसत संख्या 25 है। गुरुवार को पुस्तकालय आने वाले पाठकों की संख्या ज्ञात कीजिए।

- (A) 25
- (B) 28
- (C) 24
- (D) 30
- (E) 20

Q6: Ishita, Mohit and Kavya started a business by investing Rs. 120000, Rs. 80000 and Rs. 50000, respectively. At the end of the 8th month, Ishita and Mohit invested Rs. 40000 more each, while Kavya left the business. Find Kavya's share in the profit if the total profit at the end of the year was Rs. 46800.

इशिता, मोहित और काव्या ने क्रमशः 120000 रुपये, 80000 रुपये और 50000 रुपये निवेश करके एक व्यवसाय शुरू किया। 8वें महीने के अंत में, इशिता और मोहित ने प्रत्येक ने

40000 रुपये और निवेश किए, जबकि काव्या ने व्यवसाय छोड़ दिया। यदि वर्ष के अंत में कुल लाभ 46800 रुपये था, तो काव्या का लाभांश ज्ञात कीजिए।

- (A) Rs. 5400
- (B) Rs. 7200
- (C) Rs. 6000
- (D) Rs. 9000
- (E) Rs. 4800

Q7: Ritika and Saurabh started a business with investments of Rs. 3000 and Rs. 4000, respectively. After 5 months, Tushar joined them with an investment of Rs. 6000. Find Saurabh's share in the total profit of Rs. 6300 at the end of one year.

रितिका और सौरभ ने क्रमशः 3000 रुपये और 4000 रुपये के निवेश से एक व्यवसाय शुरू किया। 5 महीने बाद, तुषार 6000 रुपये के निवेश के साथ व्यवसाय में शामिल हुआ। एक वर्ष के अंत में 6300 रुपये के कुल लाभ में सौरभ का हिस्सा ज्ञात कीजिए।

- (A) Rs. 1800
- (B) Rs. 2700
- (C) Rs. 2100
- (D) Rs. 2400
- (E) Rs. 3000

Q8: The monthly incomes of Nivedita and Shlok are in the ratio 7 : 5, respectively, and their monthly expenses are in the ratio 5 : 3, respectively. If both of them save Rs. 900 per month, find their average monthly income.

निवेदिता और श्लोक की मासिक आय का अनुपात क्रमशः 7 : 5 है, और उनके मासिक खर्चों का अनुपात क्रमशः 5 : 3 है। यदि दोनों प्रति माह 900 रुपये बचाते हैं, तो उनकी औसत मासिक आय ज्ञात कीजिए।

- (A) Rs. 2600
- (B) Rs. 2700
- (C) Rs. 2400
- (D) Rs. 3000
- (E) Rs. 2850

Q9: A shopkeeper sells a calculator at a profit of 30%. If he had bought it at 12.5% less and sold it for Rs. 156 less, he would have gained 20%. Find the original cost price of the calculator.

एक दुकानदार एक कैलकुलेटर को 30% लाभ पर बेचता है। यदि उसने इसे 12.5% कम कीमत पर खरीदा होता और 156 रुपये कम में बेचा होता, तो उसे 20% का लाभ होता।

कैलकुलेटर का वास्तविक क्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 560
- (B) Rs. 600
- (C) Rs. 720
- (D) Rs. 624
- (E) Rs. 680

Q10: Bhavya is 12 years older than Naina, and Naina is 18 years younger than Charu. Twelve years hence, Charu's age will be 25% more than Naina's age at that time. If the average of the present ages of Naina and Manav is 50 years, find Manav's present age.

भाव्या, नैना से 12 वर्ष बड़ी है, और नैना, चारू से 18 वर्ष छोटी है। 12 वर्ष बाद, चारू की आयु उस समय नैना की आयु से 25% अधिक होगी। यदि नैना और मानव की वर्तमान आयु का औसत 50 वर्ष है, तो मानव की वर्तमान आयु ज्ञात कीजिए।

- (A) 36 years
- (B) 48 years
- (C) 52 years
- (D) 44 years
- (E) 40 years

Q11: In a mixture Y of spirit and water, the ratio of spirit to water is 7:5. When 60 litres of spirit and 120 litres of water are added to the mixture, the ratio of spirit to water becomes 5:4. Find 80% of the total quantity of the initial mixture.

मिश्रण Y में स्पिरिट और पानी का अनुपात 7:5 है। जब मिश्रण में 60 लीटर स्पिरिट और 120 लीटर पानी मिलाया जाता है, तो स्पिरिट और पानी का अनुपात 5:4 हो जाता है। प्रारंभिक मिश्रण की कुल मात्रा का 80% ज्ञात कीजिए।

- (A) 1080 litres
- (B) 1248 litres
- (C) 1152 litres
- (D) 960 litres
- (E) 1200 litres

Q12: A train running at 72 km/h crosses a cyclist moving in the same direction at 18 km/h in 16 seconds. Find the time taken by the train to cross a pole while running at the same speed.

72 किमी/घंटा की गति से चल रही एक ट्रेन समान दिशा में 18 किमी/घंटा की गति से चल रहे एक साइकिल चालक को 16 सेकंड में पार करती है। उसी गति से चलते हुए ट्रेन द्वारा एक

खंभे को पार करने में लिया गया समय ज्ञात कीजिए।

- (A) 10 seconds
- (B) 14 seconds
- (C) 12 seconds
- (D) 16 seconds
- (E) 18 seconds

Q13: The ratio of the ages of Kunal and Nitin 6 years ago was 4:5, and the ratio of their ages 6 years hence will be 7:8. If Meera's present age is 30 years, find the average of the present ages of Kunal, Nitin and Meera.

6 वर्ष पहले कुणाल और नितिन की आयु का अनुपात 4:5 था और 6 वर्ष बाद उनकी आयु का अनुपात 7:8 होगा। यदि मीरा की वर्तमान आयु 30 वर्ष है, तो कुणाल, नितिन और मीरा की वर्तमान आयु का औसत ज्ञात कीजिए।

- (A) 24 years
- (B) 28 years
- (C) 26 years
- (D) 30 years
- (E) 32 years

Q14: The monthly incomes of Naveen and Keshav are in the ratio of 5:7. If Naveen's monthly income is increased by 20%, he gets Rs. 6000 more. Find Keshav's monthly savings if his monthly expenditure is Rs. 31500.

नवीन और केशव की मासिक आय का अनुपात 5:7 है। यदि नवीन की मासिक आय में 20% की वृद्धि होती है, तो उसे Rs. 6000 अधिक मिलते हैं। यदि केशव का मासिक खर्च Rs. 31500 है, तो केशव की मासिक बचत ज्ञात कीजिए।

- (A) Rs. 10500
- (B) Rs. 11500
- (C) Rs. 9500
- (D) Rs. 12500
- (E) Rs. 10000

Q15: A retailer marked an appliance 100% above its cost price and then allowed two successive discounts of 25% and 16%, respectively. Find the profit percent earned by the retailer.

एक विक्रेता ने एक उपकरण का अंकित मूल्य उसके क्रय मूल्य से 100% अधिक रखा और फिर क्रमशः 25% और 16% की दो क्रमागत छूट दी। विक्रेता द्वारा अर्जित लाभ प्रतिशत ज्ञात कीजिए।

- (A) 20%
- (B) 24%
- (C) 26%
- (D) 28%
- (E) 30%

Q16: Kunal and Pranav invested in a partnership in the ratio $5:x$, respectively. After completing the same duration in business, Kunal received Rs 2400 from the total profit of Rs 7200. Find the value of x .

कुणाल और प्रणव ने एक साझेदारी में क्रमशः $5:x$ के अनुपात में निवेश किया। व्यवसाय में समान अवधि पूरी करने के बाद, कुल Rs 7200 के लाभ में से कुणाल को Rs 2400 प्राप्त हुए। x का मान ज्ञात कीजिए।

- (A) 8
- (B) 10
- (C) 12
- (D) 15
- (E) 6

Q17: Two numbers are in the ratio $5:9$. If the difference between the greater number and the average of both numbers is 14, find the smaller number.

दो संख्याएँ $5:9$ के अनुपात में हैं। यदि बड़ी संख्या और दोनों संख्याओं के औसत के बीच का अंतर 14 है, तो छोटी संख्या ज्ञात कीजिए।

- (A) 28
- (B) 49
- (C) 42
- (D) 35
- (E) 56

Q18: The speed of a boat in still water is three times the speed of the stream. The boat covers 48 km upstream and 96 km downstream in a total of 8 hours. Find the time taken by the boat to cover 72 km in still water.

एक नाव की स्थिर जल में गति धारा की गति की तीन गुनी है। नाव 48 किमी धारा के प्रतिकूल और 96 किमी धारा के अनुकूल कुल 8 घंटे में तय करती है। स्थिर जल में 72 किमी की दूरी तय करने में नाव द्वारा लिया गया समय ज्ञात कीजिए।

- (A) 2 hours
- (B) 3 hours
- (C) 5 hours

- (D) 6 hours
(E) 4 hours

Q19: Ritika scored 255 marks in History and failed by 45 marks. If the passing marks are 50% of the maximum marks, how many marks are required to score 75% in History?

रितिका ने इतिहास में 255 अंक प्राप्त किए और वह 45 अंकों से अनुत्तीर्ण हो गई। यदि उत्तीर्ण अंक अधिकतम अंकों के 50% हैं, तो इतिहास में 75% अंक प्राप्त करने के लिए कितने अंक चाहिए?

- (A) 450
(B) 420
(C) 480
(D) 400
(E) 525

Q20: Find the difference between the simple interest earned on Rs 7500 at 8% per annum for 4 years and the simple interest earned on Rs 9000 at 6% per annum for 3 years.

Rs 7500 पर 8% वार्षिक दर से 4 वर्षों में प्राप्त साधारण ब्याज और Rs 9000 पर 6% वार्षिक दर से 3 वर्षों में प्राप्त साधारण ब्याज के बीच का अंतर ज्ञात कीजिए।

- (A) Rs 720
(B) Rs 840
(C) Rs 780
(D) Rs 900
(E) Rs 660

Answers:

- | | |
|------|------|
| 1. D | 9. D |
| 2. E | 10.E |
| 3. C | 11.C |
| 4. B | 12.C |
| 5. A | 13.C |
| 6. C | 14.A |
| 7. D | 15.C |
| 8. B | 16.B |

17.D

19.A

18.E

20.C

Solutions:**1. Option D**

Let the principal deposited in each account = Rs. x

For the first account:

Rate = 15% p.a.

Time = 4 years

Simple interest percentage = $15 \times 4 = 60\%$

So, amount received from first account = 160% of x

Amount from first account = $x \times 160/100$

For the second account:

Rate = 10% p.a.

Time = 2 years

Compound amount = $x \times 110/100 \times 110/100$

Compound amount = $x \times 121/100$

Sum of both amounts = Rs. 8430

$x \times 160/100 + x \times 121/100 = 8430$

$x \times 281/100 = 8430$

$x = 8430 \times 100/281$

$x = 3000$

2. Option E

Ratio of radius to height = 2:3

Let radius = $2a$ and height = $3a$

Volume of cylinder = $\pi \times r \times r \times h$

$12936 = 22/7 \times 2a \times 2a \times 3a$

$12936 = 22/7 \times 12a^3$

$12936 = 264a^3/7$

$a^3 = 12936 \times 7/264$

$a^3 = 343$

$a = 7$

So, radius = $2a = 2 \times 7 = 14$ cm

Height = $3a = 3 \times 7 = 21$ cm

Curved surface area of cylinder = $2 \times \pi \times r \times h$

Curved surface area = $2 \times 22/7 \times 14 \times 21$

Curved surface area = 1848 cm^2

3. Option C

Speed of train without stoppage = 72 km/hr

Number of stoppages = 10

Time for each stoppage = 3 minutes

Total stoppage time = $10 \times 3 = 30$ minutes

30 minutes = $30/60 = 1/2$ hour

Total journey time including stoppages = 4 hours 15 minutes

4 hours 15 minutes = $4 + 15/60 = 4 + 1/4 = 17/4$ hours

Actual running time = Total time – Stoppage time

Actual running time = $17/4 - 1/2$

Actual running time = $17/4 - 2/4 = 15/4$ hours

Distance = Speed $\times$ Time

Distance = $72 \times 15/4$

Distance = $18 \times 15 = 270$ km

Upstream speed = 16 km/hr

4. Option B

For the same distance, time and speed are inversely proportional.

Time ratio of upstream to downstream = 3:2

So, speed ratio of upstream to downstream = 2:3

Let upstream speed = $2x$ and downstream speed = $3x$

$2x = 16$

$x = 8$

Downstream speed = $3x = 3 \times 8 = 24$ km/hr

Speed of stream = (Downstream speed – Upstream speed)/2

Speed of stream = $(24 - 16)/2$

Speed of stream = $8/2 = 4$ km/hr

5. Option A

Average number of readers in a week = 28

Total readers in 7 days = $28 \times 7 = 196$

Average readers on Monday, Tuesday and Wednesday = 32

Total readers on these 3 days = $32 \times 3 = 96$

Average readers on Friday, Saturday and Sunday = 25

Total readers on these 3 days = $25 \times 3 = 75$

Readers on Thursday = Total readers of week – Readers of other 6 days

Readers on Thursday = $196 - 96 - 75$

Readers on Thursday = 25

6. Option C

In partnership questions, profit is divided in the ratio of investment $\times$ time.

Ishita's capital:

For first 8 months = $120000 \times 8 = 960000$

For next 4 months = $160000 \times 4 = 640000$

Total = $960000 + 640000 = 1600000$

Mohit's capital:

For first 8 months = $80000 \times 8 = 640000$

For next 4 months = $120000 \times 4 = 480000$

Total = $640000 + 480000 = 1120000$

Kavya's capital:

Kavya invested only for 8 months.

$50000 \times 8 = 400000$

Profit ratio = $1600000 : 1120000 : 400000$

On dividing by 80000:

Profit ratio = $20 : 14 : 5$

Total parts = $20 + 14 + 5 = 39$

Kavya's share = $\frac{5}{39} \times 46800$

Kavya's share = $5 \times 1200 = \text{Rs. } 6000$

7. Option D

Profit ratio = Investment $\times$ Time

Ritika invested Rs. 3000 for 12 months.

Ritika's capital = $3000 \times 12 = 36000$

Saurabh invested Rs. 4000 for 12 months.

Saurabh's capital = $4000 \times 12 = 48000$

Tushar joined after 5 months, so he invested for 7 months.

Tushar's capital = $6000 \times 7 = 42000$

Profit ratio = $36000 : 48000 : 42000$

On dividing by 6000:

Profit ratio = $6 : 8 : 7$

$$\text{Total parts} = 6 + 8 + 7 = 21$$

$$\text{Saurabh's share} = 8/21 \times 6300$$

$$\text{Saurabh's share} = 8 \times 300 = \text{Rs. } 2400$$

8. Option B

Let the monthly incomes of Nivedita and Shlok be $7x$ and $5x$.

Let their monthly expenses be $5y$ and $3y$.

Savings = Income – Expense

Nivedita's saving:

$$7x - 5y = 900$$

Shlok's saving:

$$5x - 3y = 900$$

Subtract the second equation from the first equation:

$$7x - 5y - 5x + 3y = 900 - 900$$

$$2x - 2y = 0$$

$$\text{So, } x = y$$

Now use Shlok's saving:

$$5x - 3x = 900$$

$$2x = 900$$

$$x = 450$$

$$\text{Nivedita's income} = 7 \times 450 = \text{Rs. } 3150$$

$$\text{Shlok's income} = 5 \times 450 = \text{Rs. } 2250$$

$$\text{Average monthly income} = (3150 + 2250)/2$$

$$\text{Average monthly income} = 5400/2 = \text{Rs. } 2700$$

9. Option D

Use the banking shortcut by taking the original cost price as 8 units because $12.5\% = 1/8$.

Original cost price = 8 units

Original selling price at 30% profit = 130% of 8 units

$$\text{Original selling price} = 8 \times 130/100 = 10.4 \text{ units}$$

New cost price is 12.5% less.

$$\text{New cost price} = 8 - 1 = 7 \text{ units}$$

New selling price at 20% profit = 120% of 7 units

$$\text{New selling price} = 7 \times 120/100 = 8.4 \text{ units}$$

$$\text{Difference between both selling prices} = 10.4 - 8.4 = 2 \text{ units}$$

$$\text{Given difference} = \text{Rs. } 156$$

So,
 2 units = 156
 1 unit = 78
 Original cost price = 8 units = $8 \times 78 = \text{Rs. } 624$

10. Option E

Let Naina's present age be x years.
 Bhavya is 12 years older than Naina.
 Bhavya's present age = $x + 12$
 Naina is 18 years younger than Charu.
 Charu's present age = $x + 18$
 After 12 years:
 Naina's age = $x + 12$
 Charu's age = $x + 18 + 12 = x + 30$
 Charu's age will be 25% more than Naina's age.
 So, Charu's age = 125% of Naina's age = $\frac{5}{4}$ of Naina's age
 $x + 30 = \frac{5}{4} \times (x + 12)$
 $4x + 120 = 5x + 60$
 $x = 60$
 So, Naina's present age = 60 years.
 Average of present ages of Naina and Manav = 50 years
 Total age of Naina and Manav = $50 \times 2 = 100$ years
 Manav's present age = $100 - 60 = 40$ years

11. Option C

Let the initial quantity of spirit and water be $7x$ litres and $5x$ litres.
 After adding 60 litres of spirit, spirit = $7x + 60$
 After adding 120 litres of water, water = $5x + 120$
 New ratio = 5:4
 So, $\frac{(7x + 60)}{(5x + 120)} = \frac{5}{4}$
 $4(7x + 60) = 5(5x + 120)$
 $28x + 240 = 25x + 600$
 $3x = 360$
 $x = 120$
 Initial mixture = $7x + 5x = 12x = 12 \times 120 = 1440$ litres
 80% of initial mixture = $\frac{80}{100} \times 1440 = 1152$ litres

12. Option C

Speed of train = 72 km/h

Speed of cyclist = 18 km/h

Since both move in the same direction, relative speed = $72 - 18 = 54$ km/h

Convert 54 km/h into m/s:

$$54 \times \frac{5}{18} = 15 \text{ m/s}$$

The train crosses the cyclist in 16 seconds.

$$\text{Length of train} = 15 \times 16 = 240 \text{ meters}$$

$$\text{Speed of train} = 72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$$

Time taken to cross a pole = Length of train/Speed of train

$$\text{Time} = 240/20 = 12 \text{ seconds}$$

13. Option C

Let the present ages of Kunal and Nitin be x years and y years.

6 years ago, their ages were $x - 6$ and $y - 6$.

$$\text{So, } (x - 6)/(y - 6) = 4/5$$

$$5(x - 6) = 4(y - 6)$$

$$5x - 30 = 4y - 24$$

$$5x - 4y = 6$$

6 years hence, their ages will be $x + 6$ and $y + 6$.

$$\text{So, } (x + 6)/(y + 6) = 7/8$$

$$8(x + 6) = 7(y + 6)$$

$$8x + 48 = 7y + 42$$

$$8x - 7y = -6$$

Solving both equations:

$$5x - 4y = 6$$

$$8x - 7y = -6$$

From these, $x = 22$ and $y = 26$

Present age of Meera = 30 years

$$\text{Average age} = (22 + 26 + 30)/3$$

$$\text{Average age} = 78/3 = 26 \text{ years}$$

14. Option A

Naveen's income increases by 20%, and this increase is Rs. 6000.

$$\text{So, } 20\% \text{ of Naveen's income} = 6000$$

$$\text{Naveen's income} = 6000 \times 100/20 = \text{Rs. } 30000$$

Ratio of monthly incomes of Naveen and Keshav = 5:7

5 parts = 30000

1 part = $30000/5 = 6000$

Keshav's income = $7 \times 6000 = \text{Rs. } 42000$

Keshav's expenditure = Rs. 31500

Keshav's savings = $42000 - 31500 = \text{Rs. } 10500$

15. Option C

Let the cost price be Rs. 100.

Marked price is 100% above cost price.

Marked price = $100 + 100\% \text{ of } 100 = \text{Rs. } 200$

After 25% discount, price = 75% of 200

Price = $75/100 \times 200 = \text{Rs. } 150$

After 16% discount, final selling price = 84% of 150

Selling price = $84/100 \times 150 = \text{Rs. } 126$

Profit = $126 - 100 = \text{Rs. } 26$

Profit percent = $26/100 \times 100 = 26\%$

16. Option B

Since both partners invested for the same duration, profit will be divided in the ratio of their investments.

Kunal's profit = Rs 2400.

Total profit = Rs 7200.

Pranav's profit = $7200 - 2400 = \text{Rs } 4800$.

Profit ratio of Kunal and Pranav = $2400:4800 = 1:2$.

Investment ratio of Kunal and Pranav = $5:x$.

So, $5:x = 1:2$.

$x = 5 \times 2 = 10$.

17. Option D

Let the two numbers be $5a$ and $9a$.

Average of the two numbers = $(5a + 9a)/2 = 14a/2 = 7a$.

Difference between greater number and average = $9a - 7a = 2a$.

Given, $2a = 14$.

$a = 7$.

Smaller number = $5a = 5 \times 7 = 35$.

18. Option E

Let the speed of the stream be x km/hr.

Speed of boat in still water = $3x$ km/hr.

Upstream speed = $3x - x = 2x$ km/hr.

Downstream speed = $3x + x = 4x$ km/hr.

Total time = $48/(2x) + 96/(4x)$.

$$8 = 24/x + 24/x.$$

$$8 = 48/x.$$

$$x = 6.$$

Speed of boat in still water = $3x = 3 \times 6 = 18$ km/hr.

Time taken to cover 72 km in still water = $72 \div 18 = 4$ hours.

19. Option A

Ritika scored 255 marks and failed by 45 marks.

Passing marks = $255 + 45 = 300$.

Passing marks are 50% of maximum marks.

So, maximum marks = $300 \times 100/50 = 600$.

Marks required to score 75% = 75% of 600.

Required marks = $600 \times 75/100 = 450$.

20. Option C

Simple interest = $\text{Principal} \times \text{Rate} \times \text{Time}/100$.

First simple interest = $7500 \times 8 \times 4/100 = \text{Rs } 2400$.

Second simple interest = $9000 \times 6 \times 3/100 = \text{Rs } 1620$.

Difference = $2400 - 1620 = \text{Rs } 780$.

3. Quadratic Equations

CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $4x^2 - 52x + 168 = 0$

II. $5y^2 - 35y + 50 = 0$

2.) I. $x^2 - 9\sqrt{2}x + 28 = 0$

II. $y^2 - 10\sqrt{2}y + 32 = 0$

3.) I. $7x^2 - 7x - 84 = 0$

II. $3y^2 - 39y + 120 = 0$

4.) I. $6x^2 - 72x + 210 = 0$

II. $8y^2 - 56y + 80 = 0$

5.) I. $x^2 - 9\sqrt{6}x + 120 = 0$

II. $y^2 - 4\sqrt{6}y + 18 = 0$

6.) I. $11x^2 - 99x + 198 = 0$

II. $4y^2 - 60y + 216 = 0$

7.) I. $5x^2 - 60x + 135 = 0$

II. $4y^2 - 52y + 160 = 0$

8.) I. $x^2 - 9\sqrt{3}x + 54 = 0$

II. $y^2 - 15\sqrt{3}y + 168 = 0$

9.) I. $4x^2 - 50x + 156 = 0$

II. $7y^2 - 56y + 105 = 0$

10.) I. $3x^2 + 11x - 20 = 0$

II. $8y^2 - 40y + 48 = 0$

11.) I. $x^2 - 7\sqrt{7}x + 84 = 0$

II. $y^2 - 9\sqrt{7}y + 140 = 0$

12.) I. $7x^2 - 70x + 147 = 0$

II. $5y^2 - 60y + 175 = 0$

13.) I. $8x^2 - 60x + 112 = 0$

II. $6y^2 - 33y + 42 = 0$

14.) I. $x^2 - 10\sqrt{3}x + 72 = 0$

II. $y^2 - 4\sqrt{3}y + 9 = 0$

15.) I. $5x^2 - 20x - 60 = 0$

II. $2y^2 - 28y + 96 = 0$

16.) I. $x^2 - 10\sqrt{2}x + 42 = 0$

II. $y^2 - 11\sqrt{2}y + 48 = 0$

17.) I. $5x^2 - 49x + 114 = 0$

II. $3y^2 - 12y + 9 = 0$

18.) I. $x^2 - 7\sqrt{6}x + 72 = 0$

II. $y^2 - 11\sqrt{6}y + 180 = 0$

19.) I. $6x^2 - 6x - 72 = 0$

II. $4y^2 - 44y + 120 = 0$

20.) I. $8x^2 - 74x + 156 = 0$

II. $12y^2 - 51y + 39 = 0$

Answers:

1. A

2. C

3. B

4. D

5. A

6. E

7. C

8. B

9. A

10. B

11. E

12. C

13. D

14. A

15. E

16. C

17. A

18. B

19. B

20. D

Solutions:

$$(1) x = 7, 6$$

$$y = 5, 2$$

$$(2) x = 2\sqrt{2}, 7\sqrt{2}$$

$$y = 2\sqrt{2}, 8\sqrt{2}$$

$$(3) x = 4, -3$$

$$y = 8, 5$$

$$(4) x = 7, 5$$

$$y = 5, 2$$

$$(5) x = 4\sqrt{6}, 5\sqrt{6}$$

$$y = \sqrt{6}, 3\sqrt{6}$$

$$(6) x = 6, 3$$

$$y = 9, 6$$

$$(7) x = 9, 3$$

$$y = 8, 5$$

$$(8) x = 3\sqrt{3}, 6\sqrt{3}$$

$$y = 7\sqrt{3}, 8\sqrt{3}$$

$$(9) x = 6, \frac{13}{2}$$

$$y = 5, 3$$

$$(10) x = \frac{4}{3}, -5$$

$$y = 3, 2$$

$$(11) x = 3\sqrt{7}, 4\sqrt{7}$$

$$y = 4\sqrt{7}, 5\sqrt{7}$$

$$(12) x = 7, 3$$

$$y = 7, 5$$

$$(13) x = 4, \frac{7}{2}$$

$$y = \frac{7}{2}, 2$$

$$(14) x = 4\sqrt{3}, 6\sqrt{3}$$

$$y = \sqrt{3}, 3\sqrt{3}$$

$$(15) x = 6, -2$$

$$y = 8, 6$$

$$(16) x = 3\sqrt{2}, 7\sqrt{2}$$

$$y = 3\sqrt{2}, 8\sqrt{2}$$

$$(17) x = 6, \frac{19}{5}$$

$$y = 3, 1$$

$$(18) x = 3\sqrt{6}, 4\sqrt{6}$$

$$y = 5\sqrt{6}, 6\sqrt{6}$$

$$(19) x = 4, -3$$

$$y = 6, 5$$

$$(20) x = 6, \frac{13}{4}$$

$$y = \frac{13}{4}, 1$$

4. WRONG NUMBER SERIES

(1) 1840, 1851, 1869, 1890, 1934, 1985

- (a) 1851
- (b) 1934
- (c) 1890
- (d) 1985
- (e) None of these

(2) 6200, 6183, 6154, 6111, 6058, 5975

- (a) 6111
- (b) 6058
- (c) 6183
- (d) 5975
- (e) None of these

(3) 125, 264, 549, 1120, 2287, 4616

- (a) 264
- (b) 549
- (c) 2287
- (d) 1120
- (e) None of these

(4) 216, 633, 1886, 5647, 16935, 50789

- (a) 633
- (b) 1886
- (c) 5647
- (d) 50789
- (e) None of these

(5) 4900, 2490, 1285, 670, 360, 200

- (a) 1285
- (b) 2490
- (c) 670
- (d) 200
- (e) None of these

(6) 950, 995, 965, 1035, 980, 1055

- (a) 995
- (b) 1035
- (c) 980
- (d) 1055
- (e) None of these

(7) 1410, 1427, 1446, 1469, 1508, 1529

- | | |
|---|--|
| (a) 1427 | |
| (b) 1469 | (11) 450, 879, 1734, 3444, 6852, 13671 |
| (c) 1508 | |
| (d) 1529 | (a) 879 |
| (e) None of these | (b) 1734 |
| | (c) 6852 |
| (8) 5000, 4994, 4976, 4922, 4780, 4274 | (d) 3444 |
| (a) 4976 | (e) None of these |
| (b) 4922 | |
| (c) 4994 | (12) 125, 755, 3780, 15155, 45485, 90995 |
| (d) 4780 | |
| (e) None of these | (a) 755 |
| | (b) 3780 |
| | (c) 15155 |
| (9) 294, 366, 446, 530, 630, 734 | (d) 90995 |
| (a) 530 | (e) None of these |
| (b) 366 | |
| (c) 630 | (13) 570, 594, 634, 694, 788, 890 |
| (d) 734 | |
| (e) None of these | (a) 594 |
| | (b) 694 |
| | (c) 788 |
| (10) 3000, 2983, 2956, 2920, 2864, 2795 | (d) 890 |
| (a) 2983 | (e) None of these |
| (b) 2956 | |
| (c) 2864 | (14) 875.5, 869.25, 856.25, 838, 813, 781.75 |
| (d) 2795 | |
| (e) None of these | (a) 856.25 |
| | (b) 869.25 |

- (c) 813
(d) 781.75
(e) None of these

(15) 910, 954, 1006, 1066, 1144, 1210

- (a) 1006
(b) 954
(c) 1144
(d) 1210
(e) None of these

(16) 250.5, 262, 276.5, 294.5, 314.5, 338

- (a) 262
(b) 276.5
(c) 314.5
(d) 294.5
(e) None of these

(17) 410, 419, 448, 515, 648, 865

- (a) 419
(b) 515
(c) 648
(d) 865
(e) None of these

(18) 43348, 21669, 7208, 1795, 348, 45

- (a) 7208
(b) 21669
(c) 1795
(d) 348
(e) None of these

(19) 85, 251, 747, 2219, 6625, 19811

- (a) 251
(b) 747
(c) 6625
(d) 19811
(e) None of these

(20) 830, 842, 869, 917, 987, 1100

- (a) 842
(b) 869
(c) 917
(d) 1100
(e) None of these

Answers:

- (1) c
(2) b
(3) d
(4) e
(5) a
(6) b

(7) c	(6) +45, -30, +60, -45, +75.
(8) d	(7) +17, +19, +23, +29, +31.
(9) a	(8) -6, -18, -54, -162, -486.
(10) e	(9) $17^2 + 5$, $19^2 + 5$, $21^2 + 5$, $23^2 + 5$, $25^2 + 5$, $27^2 + 5$.
(11) d	(10) -17, -27, -39, -53, -69; second differences: -10, -12, -14, -16.
(12) b	(11) $\times 2 - 21$, $\times 2 - 24$, $\times 2 - 27$, $\times 2 - 30$, $\times 2 - 33$.
(13) c	(12) $\times 6 + 5$, $\times 5 + 10$, $\times 4 + 15$, $\times 3 + 20$, $\times 2 + 25$.
(14) a	(13) $+(4 \times 6)$, $+(5 \times 8)$, $+(6 \times 10)$, $+(7 \times 12)$, $+(8 \times 14)$.
(15) c	(14) -6.25, -12.5, -18.75, -25, -31.25.
(16) d	(15) +44, +52, +60, +68, +76; second differences: +8, +8, +8, +8.
(17) c	(16) +11.5, +14.5, +17.5, +20.5, +23.5.
(18) a	(17) $+(2^3 + 1)$, $+(3^3 + 2)$, $+(4^3 + 3)$, $+(5^3 + 4)$, $+(6^3 + 5)$.
(19) b	(18) $\div 2 - 5$, $\div 3 - 7$, $\div 4 - 9$, $\div 5 - 11$, $\div 6 - 13$.
(20) e	(19) $\times 3 - 4$, $\times 3 - 8$, $\times 3 - 16$, $\times 3 - 32$, $\times 3 - 64$.

Solutions:

- (1) +11, +18, +27, +38, +51; second differences: +7, +9, +11, +13.
- (2) -17, -29, -43, -59, -77; second differences: -12, -14, -16, -18.
- (3) $\times 2 + 14$, $\times 2 + 21$, $\times 2 + 28$, $\times 2 + 35$, $\times 2 + 42$.
- (4) $\times 3 - 15$, $\times 3 - 13$, $\times 3 - 11$, $\times 3 - 9$, $\times 3 - 7$.
- (5) $\div 2 + 40$, $\div 2 + 35$, $\div 2 + 30$, $\div 2 + 25$, $\div 2 + 20$.

- (6) +45, -30, +60, -45, +75.
- (7) +17, +19, +23, +29, +31.
- (8) -6, -18, -54, -162, -486.
- (9) $17^2 + 5$, $19^2 + 5$, $21^2 + 5$, $23^2 + 5$,
 $25^2 + 5$, $27^2 + 5$.
- (10) -17, -27, -39, -53, -69; second differences: -10, -12, -14, -16.
- (11) $\times 2 - 21$, $\times 2 - 24$, $\times 2 - 27$, $\times 2 - 30$,
 $\times 2 - 33$.
- (12) $\times 6 + 5$, $\times 5 + 10$, $\times 4 + 15$, $\times 3 + 20$,
 $\times 2 + 25$.
- (13) $+(4 \times 6)$, $+(5 \times 8)$, $+(6 \times 10)$, $+(7 \times 12)$, $+(8 \times 14)$.
- (14) -6.25, -12.5, -18.75, -25,
-31.25.
- (15) +44, +52, +60, +68, +76; second differences: +8, +8, +8, +8.
- (16) +11.5, +14.5, +17.5, +20.5, +23.5.
- (17) $+(2^3 + 1)$, $+(3^3 + 2)$, $+(4^3 + 3)$, $+(5^3 + 4)$, $+(6^3 + 5)$.
- (18) $\div 2 - 5$, $\div 3 - 7$, $\div 4 - 9$, $\div 5 - 11$, $\div 6 - 13$.
- (19) $\times 3 - 4$, $\times 3 - 8$, $\times 3 - 16$, $\times 3 - 32$,
 $\times 3 - 64$.
- (20) $+(3 \times 2^2)$, $+(3 \times 3^2)$, $+(3 \times 4^2)$, $+(3 \times 5^2)$, $+(3 \times 6^2)$.

5. MISSING NUMBER SERIES

(1) 6830, 6811, 6772, ?, 6602, 6455

- (a) 6703
- (b) 6705
- (c) 6707
- (d) 6709
- (e) None of these

(2) 7, 13, 36, ?, 688, 4119

- (a) 133
- (b) 135
- (c) 137
- (d) 139
- (e) None of these

(3) 48, 61, 87, ?, 178, 243

- (a) 126
- (b) 128
- (c) 130
- (d) 132
- (e) None of these

(4) 325, 338, 377, ?, 845, 1898

- (a) 490
- (b) 492
- (c) 494
- (d) 496
- (e) None of these

(5) 5068, 2544, ?, 220, 48, 10

(a) 852

(b) 856

(c) 860

(d) 864

(e) None of these

(6) 28.6, 33.2, 40.1, ?, 60.8, 74.6

(a) 48.9

(b) 49.1

(c) 49.3

(d) 49.5

(e) None of these

(7) 32, 45, 84, ?, 241, 404

(a) 136

(b) 140

(c) 144

(d) 148

(e) None of these

(8) 74, 88, 116, ?, 214, 284

(a) 152

(b) 154

(c) 156

(d) 158

(e) None of these

(9) 63, 128, 225, ?, 539, 768

(a) 360

(b) 364

(c) 368

- (d) 372
(e) None of these
- (10) 7800, 7775, 7675, ?, 7050, 6425
- (a) 7440
(b) 7450
(c) 7460
(d) 7470
(e) None of these
- (11) 9, 25, 81, ?, 1649, 9897
- (a) 329
(b) 331
(c) 333
(d) 335
(e) None of these
- (12) 920, 903, 852, ?, 580, 325
- (a) 744
(b) 746
(c) 748
(d) 750
(e) None of these
- (13) 138, 159, 191, ?, 296, 373
- (a) 230
(b) 232
(c) 234
(d) 236
(e) None of these
- (14) 4800, 800, 3200, ?, 2400, 240
- (a) 390
(b) 400
(c) 410
(d) 420
(e) None of these
- (15) 24, 108, 320, ?, 1512, 2744

- (a) 740
(b) 746
(c) 750
(d) 756
(e) None of these
- (16) 76, 118, 189, ?, 471, 748
- (a) 291
(b) 293
(c) 295
(d) 297
(e) None of these
- (17) 95.4, 91.8, 84.6, ?, 59.4, 41.4
- (a) 72.8
(b) 73.8
(c) 74.8
(d) 75.8
(e) None of these
- (18) 6020, 6012, 5985, ?, 5796, 5580
- (a) 5921
(b) 5923
(c) 5925
(d) 5927
(e) None of these
- (19) 64, 82, 109, ?, 226, 343
- (a) 152
(b) 154
(c) 156
(d) 158
(e) None of these
- (20) 15, 24, 44, ?, 172, 346
- (a) 82
(b) 84
(c) 86

- (d) 88
(e) None of these

Answers:

- (1) b
(2) d
(3) a
(4) c
(5) b
(6) c
(7) b
(8) d
(9) a
(10) b
(11) a
(12) d
(13) d
(14) b
(15) c
(16) d
(17) b
(18) a
(19) b
(20) c

Solutions:

- (1) $-(4^2 + 3)$, $-(6^2 + 3)$, $-(8^2 + 3)$,
 $-(10^2 + 3)$, $-(12^2 + 3)$
(2) $\times 2 - 1$, $\times 3 - 3$, $\times 4 - 5$, $\times 5 - 7$, $\times 6 - 9$
(3) $+13$, $+26$, $+39$, $+52$, $+65$
(4) $+13$, $+39$, $+117$, $+351$, $+1053$

Each difference is multiplied by 3.

$$(5) \div 2 + 10, \div 3 + 8, \div 4 + 6, \div 5 + 4, \div 6 + 2$$

$$(6) +4.6, +6.9, +9.2, +11.5, +13.8$$

$$(7) \text{ Sum of previous two numbers } +7, +11, +17, +23$$

$$(8) +14, +28, +42, +56, +70$$

$$(9) \text{ Terms are: } n^3 + 4n^2, \text{ for } n = 3, 4, 5, 6, 7, 8$$

$$(10) -5^2, -10^2, -15^2, -20^2, -25^2$$

$$(11) \times 2 + 7, \times 3 + 6, \times 4 + 5, \times 5 + 4, \times 6 + 3$$

$$(12) -17 \times 1, -17 \times 3, -17 \times 6, -17 \times 10, -17 \times 15$$

$$(13) +3 \times 7, +4 \times 8, +5 \times 9, +6 \times 10, +7 \times 11$$

$$(14) \div 6, \times 4, \div 8, \times 6, \div 10$$

$$(15) \text{ Terms are: } n^4 + n^3, \text{ for } n = 2, 3, 4, 5, 6, 7$$

$$(16) \text{ Sum of previous two numbers } -5, -10, -15, -20$$

$$(17) -3.6, -7.2, -10.8, -14.4, -18$$

$$(18) -2^3, -3^3, -4^3, -5^3, -6^3$$

$$(19) +18, +27, +45, +72, +117$$

Each difference is the sum of previous two differences.

$$(20) +9, +20, +42, +86, +174$$

Each difference = previous difference $\times 2 + 2$.

6. DATA INTERPRETATION

Set 1: The table shows the difference between the number of clay diyas and metal diyas sold by four different counters. It also shows the ratio of clay diyas to metal diyas sold by these counters. Each counter sold only these two types of diyas. **Note:** The difference is the absolute difference between clay and metal diyas.

तालिका चार अलग-अलग काउंटरों द्वारा बेचे गए मिट्टी के दीयों और धातु के दीयों की संख्या के बीच का अंतर दर्शाती है। यह इन काउंटरों द्वारा बेचे गए मिट्टी और धातु के दीयों का अनुपात भी दर्शाती है। प्रत्येक काउंटर ने केवल ये दो प्रकार के दीये बेचे।

नोट: अंतर मिट्टी और धातु के दीयों के बीच का वास्तविक अंतर है।

Counter / काउंटर	Difference between clay and metal diyas sold / बेचे गए मिट्टी और धातु के दीयों के बीच का अंतर	Ratio of clay diyas to metal diyas sold / बेचे गए मिट्टी और धातु के दीयों का अनुपात
Rewa Counter / रीवा काउंटर	44	5:7
Durg Counter / दुर्ग काउंटर	165	8:3
Hazaribagh Counter / हजारीबाग काउंटर	132	7:4
Nellore Counter / नेल्लोर काउंटर	66	3:5

Q1: The number of clay diyas sold by Hazaribagh Counter is how many more than that sold by Nellore Counter?

हजारीबाग काउंटर द्वारा बेचे गए मिट्टी के दीये, नेल्लोर काउंटर द्वारा बेचे गए मिट्टी के दीयों से कितने अधिक हैं?

- (A) 198
- (B) 220
- (C) 231
- (D) 209
- (E) 187

Q2: What is the total number of diyas sold by Hazaribagh Counter?

हजारीबाग काउंटर द्वारा बेचे गए कुल दीयों की संख्या कितनी है?

- (A) 462
- (B) 484
- (C) 506

- (D) 528
(E) 440

Q3: If 40% of the clay diya sold by Rewa Counter were painted red, then how many red clay diya were sold by Rewa Counter?

यदि रीवा काउंटर द्वारा बेचे गए मिट्टी के दीयों में से 40% लाल रंग से रंगे हुए थे, तो रीवा काउंटर ने कितने लाल मिट्टी के दीये बेचे?

- (A) 33
(B) 55
(C) 44
(D) 66
(E) 22

Q4: If each metal diya was sold by Nellore Counter for Rs. 55, then find the total amount earned by Nellore Counter by selling all metal diya.

यदि नेल्लोर काउंटर ने प्रत्येक धातु का दीया 55 रुपये में बेचा, तो सभी धातु के दीये बेचकर नेल्लोर काउंटर द्वारा अर्जित कुल राशि ज्ञात कीजिए।

- (A) Rs. 9075
(B) Rs. 8800
(C) Rs. 9350
(D) Rs. 8250
(E) Rs. 9900

Q5: If the profit earned by Durg Counter on selling each diya is Rs. 11, then find the total profit earned by Durg Counter.

यदि दुर्ग काउंटर को प्रत्येक दीया बेचने पर 11 रुपये का लाभ होता है, तो दुर्ग काउंटर द्वारा अर्जित कुल लाभ ज्ञात कीजिए।

- (A) Rs. 3883
(B) Rs. 3960
(C) Rs. 4092
(D) Rs. 3740
(E) Rs. 3993

Solutions:

Counter / काउंटर	Clay diyas / मिट्टी के दीये	Metal diyas / धातु के दीये	Total / कुल
Rewa / रीवा	110	154	264
Durg / दुर्ग	264	99	363
Hazaribagh / हजारीबाग	308	176	484
Nellore / नेल्लोर	99	165	264

Common values:

Rewa Counter: Clay:Metal = 5:7, difference = 2 parts = 44

1 part = 22

Clay = $5 \times 22 = 110$, Metal = $7 \times 22 = 154$

Durg Counter: Clay:Metal = 8:3, difference = 5 parts = 165

1 part = 33

Clay = $8 \times 33 = 264$, Metal = $3 \times 33 = 99$

Hazaribagh Counter: Clay:Metal = 7:4, difference = 3 parts = 132

1 part = 44

Clay = $7 \times 44 = 308$, Metal = $4 \times 44 = 176$

Nellore Counter: Clay:Metal = 3:5, difference = 2 parts = 66

1 part = 33

Clay = $3 \times 33 = 99$, Metal = $5 \times 33 = 165$

1. Option D

Clay diyas sold by Hazaribagh Counter = 308

Clay diyas sold by Nellore Counter = 99

Required difference = $308 - 99 = 209$

2. Option B

Total diyas sold by Hazaribagh Counter = $308 + 176 = 484$

3. Option C

Red clay diyas sold by Rewa Counter = 40% of 110

Red clay diyas = 44

4. Option A

Metal diyas sold by Nellore Counter = 165

Total amount earned = $165 \times 55 = \text{Rs. } 9075$

5. Option E

Total diyas sold by Durg Counter = $264 + 99 = 363$

Total profit = $363 \times 11 = \text{Rs. } 3993$

CHECKLIST

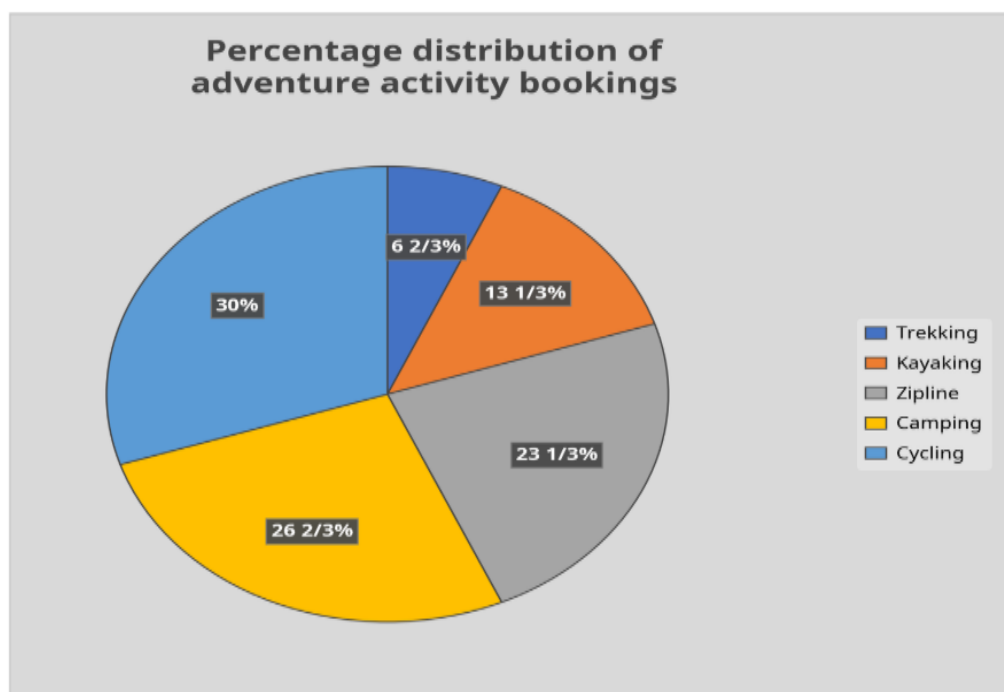
BY

AASHISH

ARORA

Set 2: The pie chart shows the percentage and degree distribution of the number of people who booked five different adventure activities at a hill camp out of the total number of people who booked activities at the camp. The total number of people who booked all five activities together is 14,400. Each person booked only one activity. Read the pie chart carefully and answer the following questions.

पाई चार्ट एक हिल कैंप में पाँच अलग-अलग एडवेंचर गतिविधियों की बुकिंग करने वाले लोगों की संख्या का प्रतिशत और डिग्री वितरण दर्शाता है। सभी पाँच गतिविधियों की कुल बुकिंग करने वाले लोगों की संख्या 14,400 है। प्रत्येक व्यक्ति ने केवल एक ही गतिविधि बुक की। पाई चार्ट को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: Find the ratio between the number of people who booked Camping to the number of people who booked Zipline.

कैंपिंग बुक करने वाले लोगों की संख्या का ज़िपलाइन बुक करने वाले लोगों की संख्या से अनुपात ज्ञात कीजिए।

- (A) 7:8
- (B) 9:7
- (C) 8:7
- (D) 5:4
- (E) 11:9

Q2: What is the average number of people who booked Trekking, Zipline and Cycling?

ट्रेकिंग, ज़िपलाइन और साइक्लिंग बुक करने वाले लोगों की औसत संख्या क्या है?

- (A) 2640
- (B) 2760
- (C) 3000
- (D) 3120
- (E) 2880

Q3: Out of the total number of people who booked Camping and Kayaking, 62.5% and 75%, respectively, paid the booking amount online. Find the total number of people who booked Camping or Kayaking but did not pay online.

कैंपिंग और कयाकिंग बुक करने वाले कुल लोगों में से क्रमशः 62.5% और 75% लोगों ने बुकिंग राशि ऑनलाइन चुकाई। कैंपिंग या कयाकिंग बुक करने वाले लेकिन ऑनलाइन भुगतान न करने वाले लोगों की कुल संख्या ज्ञात कीजिए।

- (A) 1920
- (B) 1840
- (C) 2040
- (D) 2160
- (E) 1680

Q4: The ratio of men, women and children who booked Camping is 5:7:12, respectively. The number of people who booked Cycling is how much percentage more than the number of children who booked Camping?

कैंपिंग बुक करने वाले पुरुषों, महिलाओं और बच्चों का अनुपात क्रमशः 5:7:12 है।

साइक्लिंग बुक करने वाले लोगों की संख्या, कैंपिंग बुक करने वाले बच्चों की संख्या से कितने प्रतिशत अधिक है?

- (A) 100%
- (B) 150%
- (C) 175%
- (D) 125%
- (E) 200%

Q5: Find the difference between the number of people who booked Cycling and the number of people who booked Kayaking.

साइक्लिंग बुक करने वाले लोगों और कयाकिंग बुक करने वाले लोगों की संख्या के बीच का अंतर ज्ञात कीजिए।

- (A) 2160
 (B) 2400
 (C) 2280
 (D) 2520
 (E) 2640

Solutions:

Activity	Percentage	Central angle	Number of people
Trekking	6 2/3%	24°	960
Kayaking	13 1/3%	48°	1920
Zipline	23 1/3%	84°	3360
Camping	26 2/3%	96°	3840
Cycling	30%	108°	4320
Total	100%	360°	14400

Common Calculation:

Total number of people = 14400

Trekking = 6 2/3% of 14400 = 960

Kayaking = 13 1/3% of 14400 = 1920

Zipline = 23 1/3% of 14400 = 3360

Camping = 26 2/3% of 14400 = 3840

Cycling = 30% of 14400 = 4320

1. Option C

Camping bookings = 3840

Zipline bookings = 3360

Required ratio = 3840:3360 = 8:7

2. Option E

Trekking bookings = 960

Zipline bookings = 3360

Cycling bookings = 4320

Required average = $(960 + 3360 + 4320)/3 = 2880$

3. Option A

Camping bookings = 3840

People who did not pay online for Camping = 37.5% of 3840 = 1440

Kayaking bookings = 1920

People who did not pay online for Kayaking = 25% of 1920 = 480

Required total = 1440 + 480 = 1920

4. Option D

Camping bookings = 3840

Men : Women : Children = 5:7:12

Children who booked Camping = $3840 \times \frac{12}{24} = 1920$

Cycling bookings = 4320

Difference = 4320 - 1920 = 2400

Required percentage = $2400/1920 \times 100 = 125\%$

5. Option B

Cycling bookings = 4320

Kayaking bookings = 1920

Required difference = 4320 - 1920 = 2400

Set 3: Three types of practice booklets, Maths, Reasoning and English, are ordered from an online learning portal on different days of a week. The table shows the percentage of Maths booklets ordered out of the total number of booklets ordered till a particular day and the ratio of Reasoning booklets to English booklets ordered till that day. Read the data carefully and answer the following questions.

एक ऑनलाइन लर्निंग पोर्टल से सप्ताह के अलग-अलग दिनों में तीन प्रकार की प्रैक्टिस बुकलेट्स, Maths, Reasoning और English, ऑर्डर की गईं। तालिका किसी विशेष दिन तक ऑर्डर की गई कुल बुकलेट्स में Maths बुकलेट्स का प्रतिशत और उस दिन तक ऑर्डर की गई Reasoning बुकलेट्स तथा English बुकलेट्स का अनुपात दर्शाती है। डेटा को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।

Day (दिन)	Total booklets ordered till that day (उस दिन तक कुल बुकलेट्स)	Maths booklets % (बुकलेट्स %)	Reasoning : English (Reasoning : English)
Monday (सोमवार)	840	40%	5 : 4
Till Tuesday (मंगलवार तक)	1600	36%	5 : 3
Till Wednesday (बुधवार तक)	2400	32%	8 : 9
Till Thursday (गुरुवार तक)	3200	28%	7 : 9
Till Friday (शुक्रवार तक)	4000	32.5%	7 : 8

Q1: If the ratio of the number of Reasoning booklets to English booklets ordered only on Thursday is $p : q$, then find the value of $(p + q)$.

यदि केवल गुरुवार को ऑर्डर की गई Reasoning बुकलेट्स और English बुकलेट्स का अनुपात $p : q$ है, तो $(p + q)$ का मान ज्ञात कीजिए।

- (A) 11
- (B) 12
- (C) 13
- (D) 14
- (E) 15

Q2: The number of Maths booklets ordered on Tuesday is what percent more or less than the number of Maths booklets ordered on Wednesday?

मंगलवार को ऑर्डर की गई Maths बुकलेट्स की संख्या, बुधवार को ऑर्डर की गई Maths बुकलेट्स की संख्या से कितने प्रतिशत अधिक या कम है?

- (A) 20% more
- (B) 25% more

- (C) 16.66% less
- (D) 12.5% more
- (E) None of these

Q3: Find the average number of English booklets ordered on Tuesday and Friday.
मंगलवार और शुक्रवार को ऑर्डर की गई English बुकलेट्स की औसत संख्या ज्ञात कीजिए।

- (A) 148
- (B) 156
- (C) 144
- (D) 160
- (E) 152

Q4: Find the difference between the number of English booklets ordered on Wednesday and Thursday.

बुधवार और गुरुवार को ऑर्डर की गई English बुकलेट्स की संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 36
- (B) 44
- (C) 48
- (D) 52
- (E) 40

Q5: If the total number of booklets ordered on Wednesday, Thursday and Friday together is represented in a pie chart, then find the central angle made by the number of booklets ordered on Friday.

यदि बुधवार, गुरुवार और शुक्रवार को मिलाकर ऑर्डर की गई कुल बुकलेट्स को पाई चार्ट में दर्शाया जाए, तो शुक्रवार को ऑर्डर की गई बुकलेट्स द्वारा बनाया गया केंद्रीय कोण ज्ञात कीजिए।

- (A) 120
- (B) 135
- (C) 144
- (D) 150
- (E) 108

Solutions:

Day (दिन)	Maths	Reasoning	English	Total
Monday (सोमवार)	336	280	224	840
Tuesday (मंगलवार)	240	360	160	760
Wednesday (बुधवार)	192	128	480	800
Thursday (गुरुवार)	128	240	432	800
Friday (शुक्रवार)	404	252	144	800

Common calculation:

Till Monday: Maths = 40% of 840 = 336, Reasoning : English = 5 : 4, so Reasoning = 280 and English = 224

Till Tuesday: Maths = 36% of 1600 = 576, Reasoning : English = 5 : 3, so Reasoning = 640 and English = 384

Till Wednesday: Maths = 32% of 2400 = 768, Reasoning : English = 8 : 9, so Reasoning = 768 and English = 864

Till Thursday: Maths = 28% of 3200 = 896, Reasoning : English = 7 : 9, so Reasoning = 1008 and English = 1296

Till Friday: Maths = 32.5% of 4000 = 1300, Reasoning : English = 7 : 8, so Reasoning = 1260 and English = 1440

Daily data:

Monday: Maths = 336, Reasoning = 280, English = 224, Total = 840

Tuesday: Maths = 576 - 336 = 240, Reasoning = 640 - 280 = 360, English = 384 - 224 = 160, Total = 760

Wednesday: Maths = 768 - 576 = 192, Reasoning = 768 - 640 = 128, English = 864 - 384 = 480, Total = 800

Thursday: Maths = 896 - 768 = 128, Reasoning = 1008 - 768 = 240, English = 1296 - 864 = 432, Total = 800

Friday: Maths = 1300 - 896 = 404, Reasoning = 1260 - 1008 = 252, English = 1440 - 1296 = 144, Total = 800

1. Option D

Reasoning booklets ordered on Thursday = 240

English booklets ordered on Thursday = 432

Required ratio = 240 : 432 = 5 : 9

$$p + q = 5 + 9 = 14$$

2. Option B

Maths booklets ordered on Tuesday = 240

Maths booklets ordered on Wednesday = 192

Required percentage = $(240 - 192) / 192 \times 100$

Required percentage = $48 / 192 \times 100 = 25\%$ more

3. Option E

English booklets ordered on Tuesday = 160

English booklets ordered on Friday = 144

Average = $(160 + 144) / 2$

Average = 152

4. Option C

English booklets ordered on Wednesday = 480

English booklets ordered on Thursday = 432

Difference = $480 - 432 = 48$

5. Option A

Total booklets ordered on Wednesday, Thursday and Friday = $800 + 800 + 800 = 2400$

Booklets ordered on Friday = 800

Central angle = $800 / 2400 \times 360$

Central angle = 120

Set 4: Three tour guides, Aarav, Bhavesh and Chaitanya, conducted two types of walks: Heritage Walks and Food Walks. Aarav conducted 260 Heritage Walks, which is 25% more than the Heritage Walks conducted by Bhavesh. The total number of walks conducted by Bhavesh is 104 more than the number of Heritage Walks conducted by Aarav. The ratio of Food Walks conducted by Bhavesh to Food Walks conducted by Chaitanya is 6 : 5. The number of Food Walks conducted by Chaitanya is 52 more than the number of Heritage Walks conducted by him. The total number of walks conducted by Chaitanya is 104 less than the total number of walks conducted by Aarav.

तीन टूर गाइड, आरव, भावेश और चैतन्य ने दो प्रकार की वॉक, धरोहर वॉक और फूड वॉक, कराई। आरव ने 260 धरोहर वॉक कराई, जो भावेश द्वारा कराई गई धरोहर वॉक से 25% अधिक हैं। भावेश द्वारा कराई गई कुल वॉक, आरव द्वारा कराई गई धरोहर वॉक से 104 अधिक हैं। भावेश और चैतन्य द्वारा कराई गई फूड वॉक का अनुपात 6 : 5 है। चैतन्य द्वारा कराई गई फूड वॉक की संख्या, उसके द्वारा कराई गई धरोहर वॉक से 52 अधिक है। चैतन्य द्वारा कराई गई कुल वॉक, आरव द्वारा कराई गई कुल वॉक से 104 कम है।

Q1: Find the difference between the total number of walks conducted by Bhavesh and the number of Heritage Walks conducted by Chaitanya.

भावेश द्वारा कराई गई कुल वॉक और चैतन्य द्वारा कराई गई धरोहर वॉक की संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 273
- (B) 299
- (C) 286
- (D) 312
- (E) 260

Q2: The number of Heritage Walks conducted by Bhavesh is what percentage of the number of Food Walks conducted by Aarav?

भावेश द्वारा कराई गई धरोहर वॉक की संख्या, आरव द्वारा कराई गई फूड वॉक की संख्या का कितना प्रतिशत है?

- (A) 300%
- (B) 350%
- (C) 450%
- (D) 375%
- (E) 400%

Q3: Find the average number of Heritage Walks conducted by Aarav, Bhavesh and Chaitanya.

आरव, भावेश और चैतन्य द्वारा कराई गई धरोहर वॉक की औसत संख्या ज्ञात कीजिए।

- (A) 182
- (B) 195
- (C) 169
- (D) 208
- (E) 176

Q4: If Ishaan conducted Heritage Walks that are 50% more than the average number of Heritage Walks conducted by Aarav and Bhavesh, then Ishaan's Heritage Walks are what percentage more than the Food Walks conducted by Aarav?

यदि ईशान द्वारा कराई गई धरोहर वॉक की संख्या, आरव और भावेश द्वारा कराई गई धरोहर वॉक की औसत संख्या से 50% अधिक है, तो ईशान की धरोहर वॉक, आरव द्वारा कराई गई फूड वॉक से कितने प्रतिशत अधिक है?

- (A) 525%
- (B) 625%
- (C) 550%
- (D) 575%
- (E) 600%

Q5: Find the total number of walks conducted by Aarav, Bhavesh and Chaitanya together.

आरव, भावेश और चैतन्य द्वारा मिलाकर कराई गई कुल वॉक की संख्या ज्ञात कीजिए।

- (A) 858
- (B) 884
- (C) 910
- (D) 871
- (E) 897

Solutions:

Guide	Heritage Walks	Food Walks	Total Walks
Aarav	260	52	312
Bhavesh	208	156	364
Chaitanya	78	130	208
Total	546	338	884

1. Option C

Total walks conducted by Bhavesh = 364

Heritage Walks conducted by Chaitanya = 78

Required difference = $364 - 78$

= 286

2. Option E

Heritage Walks conducted by Bhavesh = 208

Food Walks conducted by Aarav = 52

Required percentage = $208 \div 52 \times 100$

= 400%

3. Option A

Heritage Walks by Aarav, Bhavesh and Chaitanya = 260, 208 and 78

Required average = $(260 + 208 + 78) \div 3$

= $546 \div 3$

= 182

4. Option D

Average Heritage Walks of Aarav and Bhavesh = $(260 + 208) \div 2$

= 234

Ishaan's Heritage Walks = 150% of 234

= 351

Food Walks conducted by Aarav = 52

Required percentage more = $(351 - 52) \div 52 \times 100$

= 575%

5. Option B

Total walks conducted by Aarav = 312

Total walks conducted by Bhavesh = 364

Total walks conducted by Chaitanya = 208

Required total = $312 + 364 + 208$

= 884